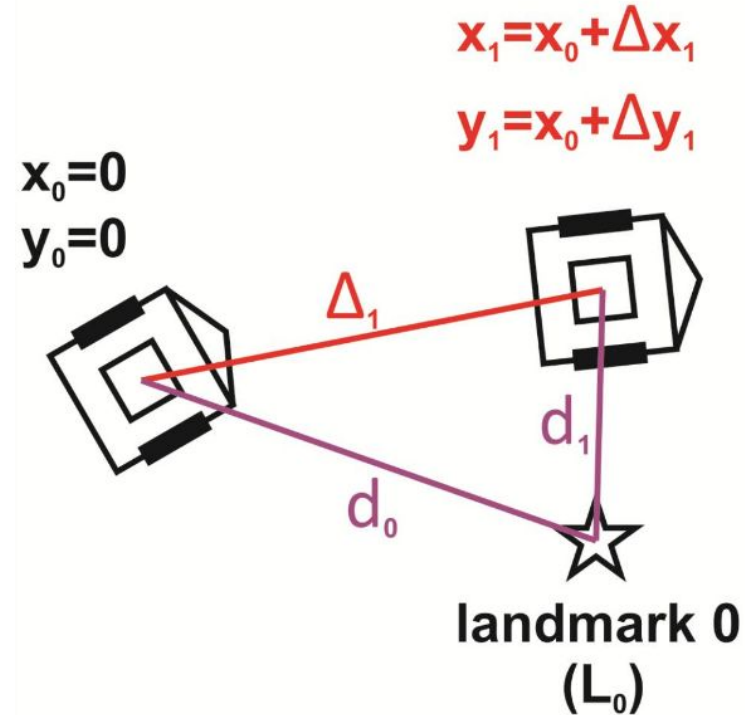


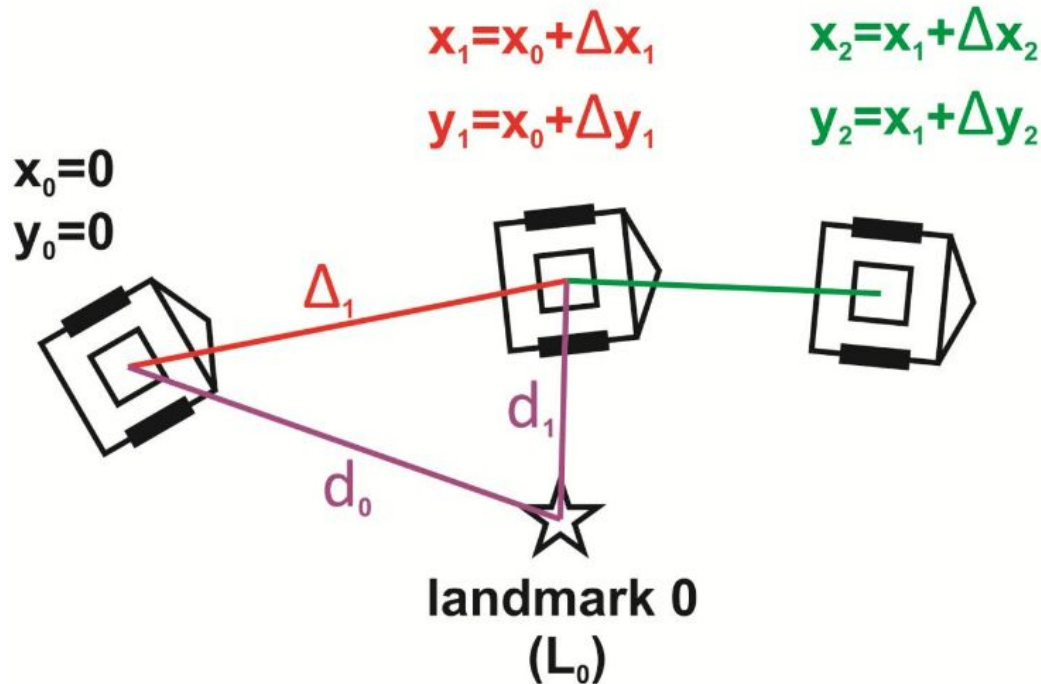
Graph-based SLAM - what is ?

- A technique that uses a graph structure to solve the problem of Simultaneous Localization and Mapping (SLAM)
- Each node corresponds to a pose (position + orientation) of the robot or landmark
- Each edge corresponds to a spatial constraint between two nodes, derived from sensor measurements, *e.g.*, odometry between consecutive positions (Δ_1) or distance measured between robot and landmark (d_0)
- The goal is to minimize the total error across the graph (*e.g.*, accumulated drift from odometry)



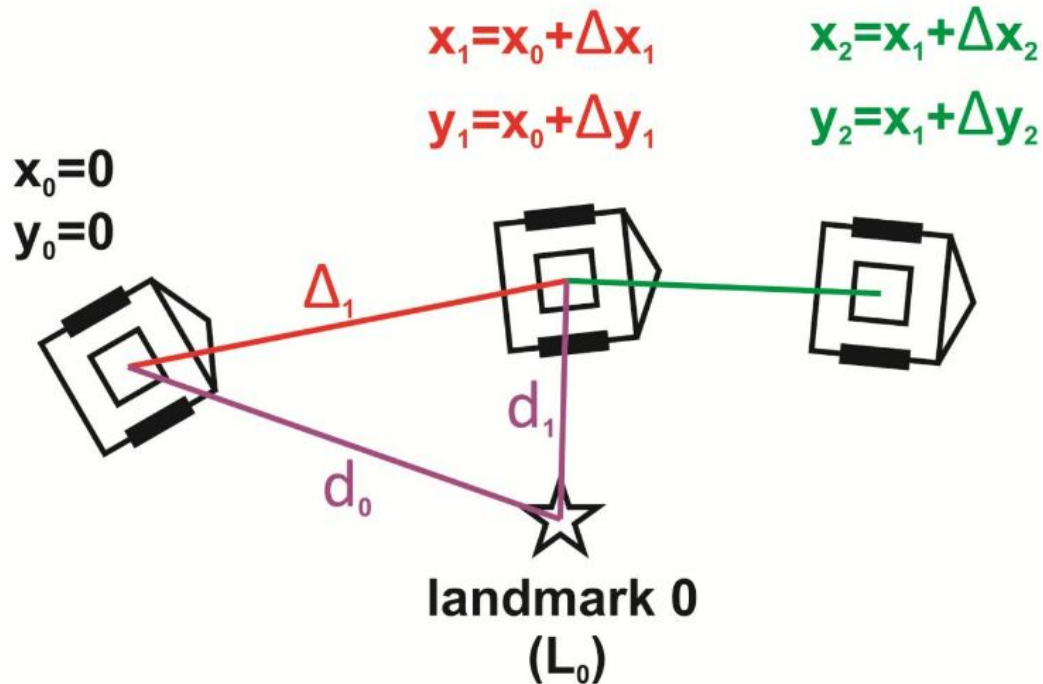
Graph-based SLAM - what is ?

- The robot keeps moving and reaches a new position (x_2, y_2) . This position is calculated using wheel odometry $(\Delta x_2, \Delta y_2)$



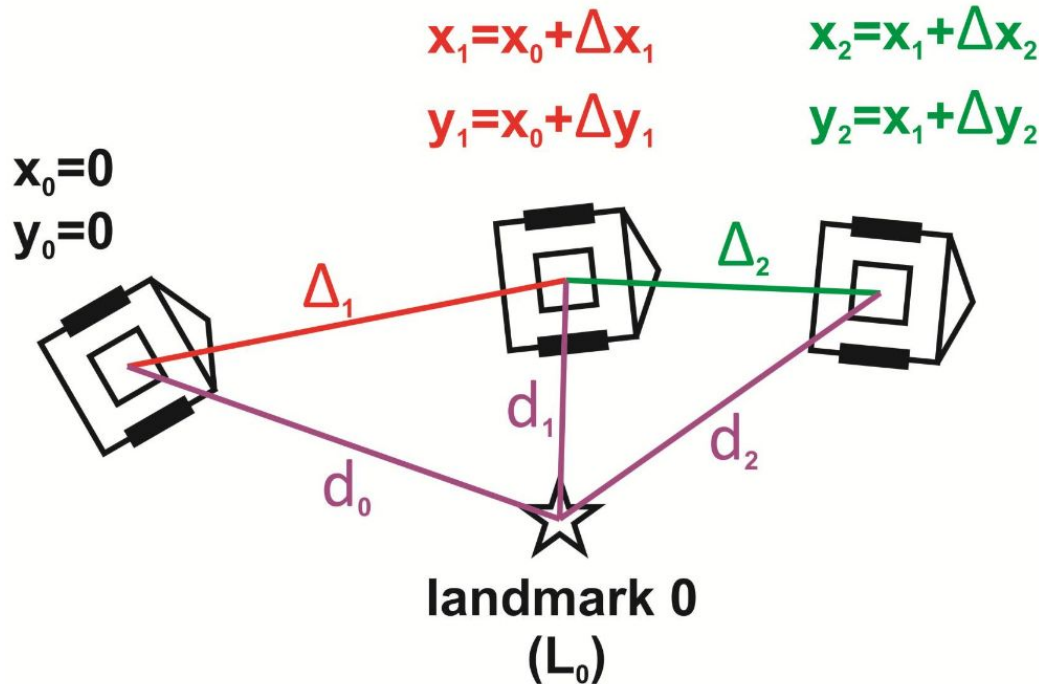
Graph-based SLAM - what is ?

- However, wheel odometry is noisy, so it carries uncertainty for position calculation



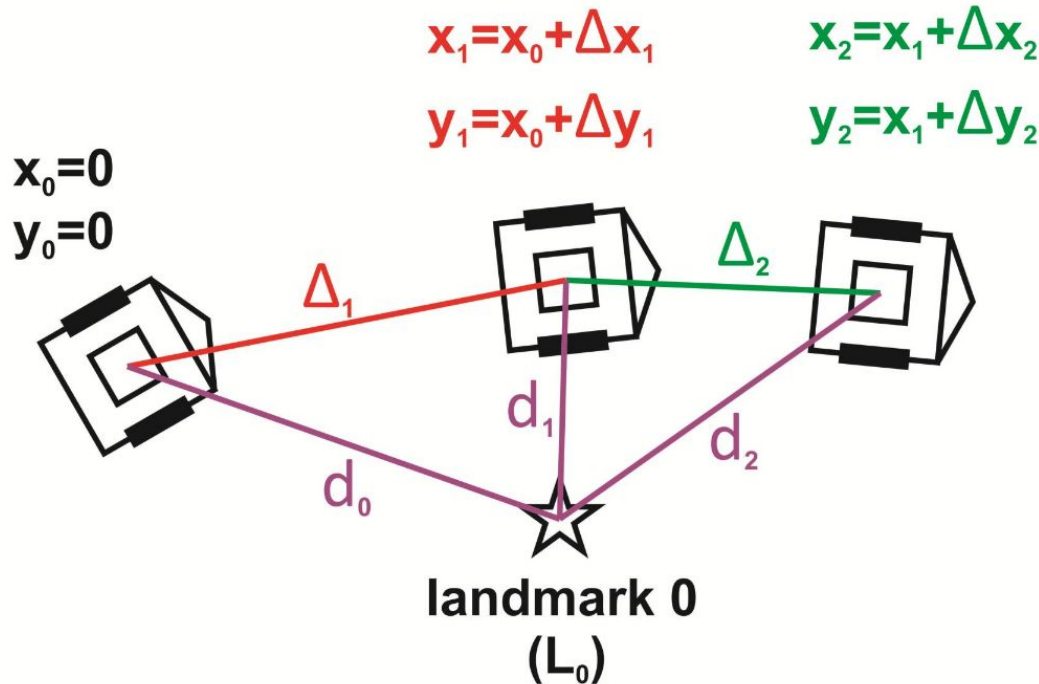
Graph-based SLAM - what is ?

- Using a reliable sensor, the robot detects the landmark L_0 again, measuring a distance d_2 between the robot and the landmark



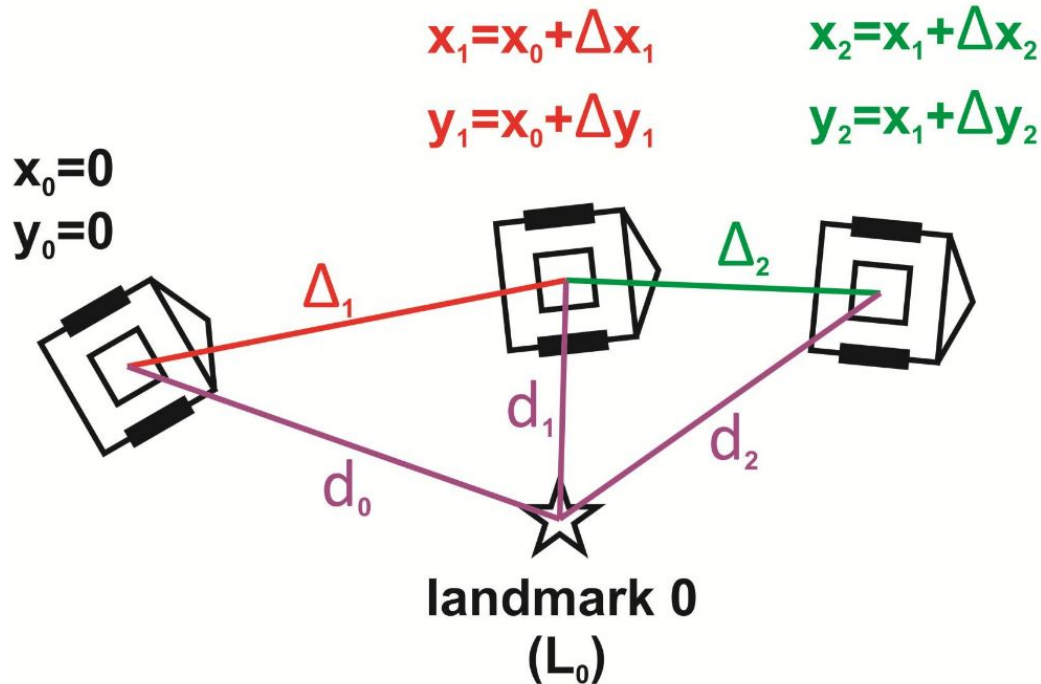
Graph-based SLAM - what is ?

- Usually, if Δ_2 is noisy, the position calculated using d_2 too is different from the position calculated using only Δ_2 , highlighting drift from wheel odometry



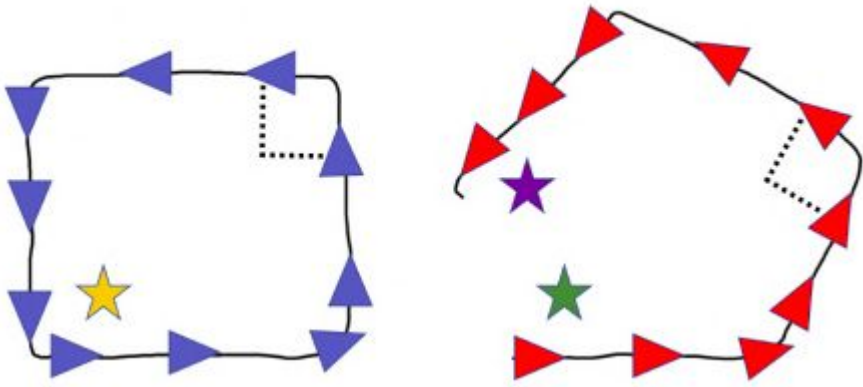
Graph-based SLAM - what is ?

- Therefore, accumulated errors can be corrected using graph-based SLAM technique

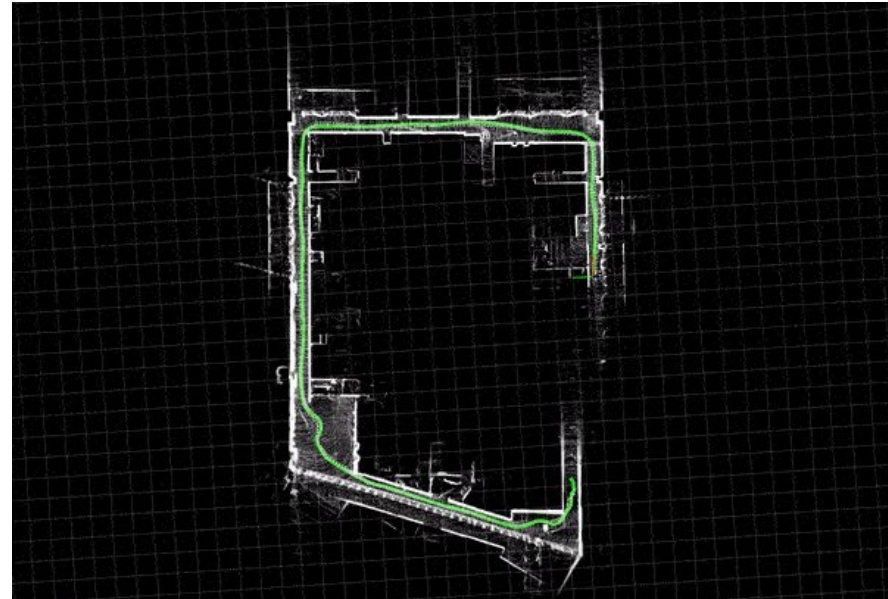


Graph-based SLAM - loop closure

- Besides the pose graph being used to compute robot pose, it also finds loop closures
- If a loop closure is found, the pose graph is optimized, and pose estimates are updated



The same landmark is spotted in both images, but loop closure is only enabled in the blue graph.



Graph-based SLAM - classes from Udacity

- Quiz: GraphSLAM: <https://www.youtube.com/watch?v=nLEbJZFm5-E> total: 3m50
A: <https://www.youtube.com/watch?v=8m4wsC8qaS0>
- Quiz: Implementing Constraints: <https://www.youtube.com/watch?v=V41eTIGU0gw>
A: <https://www.youtube.com/watch?v=D3lnErHAehY>
- Quiz: Adding Landmarks: <https://www.youtube.com/watch?v=esO-DoNEamA>
A: <https://www.youtube.com/watch?v=KNSV0Juk2YY>
- Quiz: SLAM Quiz: <https://www.youtube.com/watch?v=QF-CHsUIRUE>
A: <https://www.youtube.com/watch?v=drk0PFtoec>
- Quiz: Matrix Modification: <https://www.youtube.com/watch?v=cFObs2UufNE>
A: <https://www.youtube.com/watch?v=0wubvPa5CdU>
- Quiz: Untouched Fields: <https://www.youtube.com/watch?v=azLf8SEIM-M>
A: <https://www.youtube.com/watch?v=PGqgn2Pellc>
- Wrap up: Omega and Xi <https://www.youtube.com/watch?v=U8iFMhtJyek> until 1min 35

SLAM Toolbox

- Example of graph-based SLAM algorithm, SLAM Toolbox is also the currently supported ROS2-SLAM library
- The original repository: https://github.com/SteveMacenski/slam_toolbox
- Let's start working with this package, but let's continue in Codeshare (link available on Moodle)

Thank You...

...for your attention

